JAKHONGIR NODIROV

Robotics / Computer Vision Engineer

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SUMMARY

Computer Vision / AI engineer with hands-on robotics experience – ROS2 navigation and SLAM, sensor integration (LiDAR, encoders, motor control), and embedded deployment on Jetson and microcontroller platforms. Full-cycle experience from perception model research through real-time edge inference, including CNN-based defect inspection models running on production AOI lines. Recent work covers robot learning with LeRobot and deploying a vision-language-action model on a Jetson. Author of 2 peer-reviewed publications on 3D medical imaging. Comfortable across perception, embedded control, and on-device optimization.

TECHNICAL SKILLS

- **Robotics:** ROS2, MicroROS, SLAM, Nav2, sensor integration (LiDAR, encoders, motor drivers), PID motor control
- **Robot Learning:** LeRobot, behavior cloning, PPO (Stable-Baselines3), MuJoCo
- **Embedded/Edge:** Jetson Nano, Jetson Xavier NX, Raspberry Pi, ESP32, TensorRT, ONNX, ONNX Runtime (C++), Yocto, PlatformIO, Arduino/C++
- **Perception & ML:** PyTorch, PyTorch3D, TensorFlow, OpenCV, Object Detection/Segmentation/Tracking, Pose & Landmark Estimation
- **Languages:** Python, C/C++
- **Deployment/Infra:** Docker/Compose, AWS EC2/S3, Oracle Cloud, CI/CD

WORK EXPERIENCE

HyVision System Inc. Seongnam, South Korea
AI Research Software Engineer Jun 2025 – Sep 2026

- Built CNN-based defect detection and segmentation models for automated optical inspection (AOI) equipment, owning each from labeling rules and dataset building through training and TensorRT optimization to integration in the C++ inspection software running on production lines.
- Delivered models for two product lines: cosmetic inspection of iPhone enclosure parts for Apple through a tier-1 supplier, and semiconductor-wafer inspection.
- Set up multi-GPU training on the team's new training server.

AIVAR Seoul, South Korea
AI Research Software Engineer Feb 2023 – Mar 2025

- Led development of a 3D human avatar reconstruction server application (3D morphable face/body models), deployed on AWS with Docker for consistent dev-to-production environments.
- Built and shipped a deep-learning image verification system (vector embedding search) for animation-character and logo protection in a government-backed digital IP-protection initiative.
- Researched and optimized CNN- and transformer-based vision models across multiple concurrent product lines; owned deployment and production monitoring.

Gachon University – CT Lab Seongnam, South Korea
Graduate Researcher Sep 2020 – Feb 2023

- Designed a novel attention-based 3D U-Net with multiple skip connections for volumetric MRI tumor segmentation, reaching whole-tumor Dice 0.8974 on BraTS-2020; published in Sensors (2022), cited 130+ times.
- Built VTK-based 3D volume reconstruction from MRI slices, published at IEEE ICTC (2021).
- Contributed to applied CV lab projects including road-condition assessment and license-plate recognition.

SELECTED PROJECTS

iPhone Enclosure Cosmetic Inspection *HyVision for Apple, 2025-2026*

- Built the multi-stage inspection pipeline for 120-megapixel scans of iPhone enclosure parts, running on production AOI machines at Apple's tier-1 supplier. A segmentation model finds the part, a lightweight CNN detector screens it tile by tile for more than a dozen defect classes, and CNN segmentation and classification models outline and label each defect. Set labeling rules from the cosmetic specification, wrote multi-class auto-labeling and missed-label mining tools, and verified that Python and C++ TensorRT outputs matched.
- PyTorch, ONNX, TensorRT, C++ · Raised detection mAP50 and recall by about a quarter each through rare-class sample generation, class balancing and specification-aligned labels, and the detector matched an off-the-shelf detector's accuracy at 1.6× its speed. A multi-threaded TensorRT runner made full-scan detection 8× faster, within the line's cycle time, and moving the segmentation model to TensorRT made it about 100× faster.

Semiconductor Wafer Defect Inspection *HyVision, 2025-2026*

- Developed the CNN detectors for active-area and whitening defects that run in the C++ inspection software on production equipment, building grayscale training sets from mask labels and model-assisted auto-labeling. Retraining on merged old and new data raised active-area mAP50 by about 12%, and PyTorch and ONNX outputs were verified to match.
- PyTorch, ONNX, TensorRT, C++ · Designed a lightweight detector with a third of the baseline's parameters, trained on more than 50k tiles, plus a parallel, dynamically batched TensorRT runner. Together they reached about 11× the throughput of tile-by-tile inference with mAP50 within 3%, and the lightweight detector went on to power the iPhone enclosure inspection.

Autonomous Driving Robot (ROS2) *Side Project*

- Designed and built a mobile robot and its Gazebo twin running one ROS2 stack: a Jetson Xavier NX on ROS2 Humble in Docker (first built on a Jetson Nano B01 with ROS2 Foxy), an ESP32-S3 driving TB6612 motor drivers and streaming encoder and IMU data at 50 Hz, an RP-Lidar for range sensing, and an ESP32-CAM for vision.
- slam_toolbox builds maps from the real lidar, and AMCL localization and Nav2 navigate the Gazebo twin, built from measurements of the chassis; an EKF fuses wheel odometry and IMU, and a web dashboard handles teleoperation (roslib.js). Demo video

Camera-Only Docking *Side Project, 2026*

- Built a learning-from-demonstration stack for a ROS2 robot: teleoperated demonstrations recorded as a LeRobot v3 dataset, a ResNet18 behavior-cloning policy exported to ONNX and run in a ROS2 node, a C++ safety gate as the only path to the motors, and a referee node that scores every method in pre-registered paired trials against a classical servo. Tested in simulation.

SmolVLA Edge *Side Project, 2026*

- Exported the SmolVLA vision-language-action policy as three ONNX graphs driven by a C++17 runtime, reproducing LeRobot's actions to 6e-6 in FP32 on the real LIBERO checkpoint, and ran them on TensorRT on a Jetson Xavier NX: one action chunk in 1.0 s at FP32 and 0.33 s at FP16, with FP16 actions within the range of LeRobot's own BF16.

Driver Monitoring System (DMS) *Lab Project*

- Built a driver-monitoring pipeline on Jetson Xavier NX with nine TensorRT FP16 engines for face detection, landmarks, head pose, and eye and mouth state, plus alert logic, running end to end at 16–30 FPS. Tested on recorded driving video.

Seamline *Open Source, 2026*

- Built a desktop app for segmentation on Electron and PyTorch: label images, train a model, review its proposals least certain first, and retrain on what was accepted. Remote training runs detached over SSH and survives lost connections; large images open through tile pyramids.
- Built on mySegLib, a YAML-driven PyTorch segmentation library with DDP multi-GPU training, mixed precision, chunked inference for large images, and ONNX export. CI runs the full pipeline on synthetic data.

3D Human Avatar Reconstruction *AIVAR, 2023-2024*

- Converted 2D images into textured 3D face/body avatars using 3D morphable face and body models, trained and customized for Korean faces and bodies, with diffusion-based image correction and neural texture-matching.

EDUCATION

Gachon University 2020.09 – 2022.08
Master's Degree, Computer Engineering – GPA 4.17/4.5

PUBLICATIONS

- "Attention 3D U-Net with Multiple Skip Connections for Segmentation of Brain Tumor Images." *Sensors* 22, no. 17 (2022): 6501.
- "3D Volume Reconstruction from MRI Slices based on VTK." In *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 689–692. IEEE, 2021.

LANGUAGES

Uzbek (Native) · English (Proficient) · Korean (Intermediate)